

Lecture 11: Steepest Descent and Newton's Methods

Instructor Proof Notes

Color Legend

Green = Proof steps and logical progression

Orange = Key definitions and core concepts

Purple = Main conclusions and results

Blue boxes = Teaching notes and board diagrams

Red boxes = Common student confusions

Proof Roadmap

Theorem	Main Topic	Time (min)	Priority
Theorem 1	Steepest Descent Direction	5–7	High
Theorem 2	SD Convergence (Exact LS)	8–10	High
Theorem 3	QP Convergence Rate	6–8	Medium
Theorem 4	Newton Direction Descent	5–6	High
Theorem 5	Newton Quadratic (1-step)	4–5	High
Theorem 6	Quadratic Convergence (local)	6–8	Medium

Total estimated lecture time: 34–44 minutes. Adjust based on student engagement and prior knowledge.

1 Theorem 1: Steepest Descent Direction Lemma

Statement

The direction $d^* = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$ solves the minimization problem:

$$d^* = \arg \min_{\|d\| \leq 1} \nabla f(x_k)^T d$$

Proof

Step 1: Problem Setup: We seek to minimize the directional derivative $\nabla f(x_k)^T d$ over all unit-norm directions. This asks: which direction causes the steepest decrease in f ?

Step 2: Apply Cauchy-Schwarz: For any vectors $u, v \in \mathbb{R}^n$:

$$u^T v \geq -\|u\| \cdot \|v\|$$

with equality when $v = -cu$ for some $c > 0$.

Step 3: Apply to our problem: Let $u = \nabla f(x_k)$ and $v = d$ with $\|d\| \leq 1$:

$$\nabla f(x_k)^T d \geq -\|\nabla f(x_k)\| \cdot \|d\| \geq -\|\nabla f(x_k)\|$$

The minimum value $-\|\nabla f(x_k)\|$ is achieved when $\|d\| = 1$ and $d = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$.

The steepest descent direction is $d^* = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$, and the rate of decrease is $-\|\nabla f(x_k)\|$.

Teaching Notes

Board Diagram

Draw on board:

- Unit ball centered at x_k in $\mathbb{R}^2$ (or $\mathbb{R}^3$)
- Gradient vector $\nabla f(x_k)$ pointing outward
- Level curve (or level set) of f passing through x_k
- Optimal direction d^* pointing opposite to $\nabla f(x_k)$ on the circle boundary
- Label: “Steepest *decrease*” along $-\nabla f(x_k)$ direction

Pedagogical comment: Emphasize that we’re minimizing the directional derivative, not maximizing. The negative sign is crucial!

Common Confusion

Student Mistake: “Why do we use the unit direction instead of just $\nabla f(x_k)$ itself?”

Answer: The unit direction decouples the direction of descent from the magnitude. We normalize to make the problem well-defined (the gradient itself has arbitrary magnitude). In practice, we’ll scale the direction with a step size α_k in the actual update.

2 Theorem 2: Convergence of Steepest Descent (Exact Line Search)

Statement

Suppose at each iteration we perform an **exact line search**:

$$\alpha_k = \arg \min_{\alpha \geq 0} f(x_k + \alpha d_k), \quad d_k = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$$

If the sequence $\{x_k\}$ converges to x^* , then $\nabla f(x^*) = 0$.

Proof

Step 1: Define the directional derivative along the descent direction:

$$\theta_k(\alpha) := f(x_k + \alpha d_k)$$

where $d_k = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$ (unit direction).

Step 2: Exact line search condition: The optimal step size α_k satisfies $\theta'_k(\alpha_k) = 0$, which means:

$$\theta'_k(\alpha_k) = \nabla f(x_k + \alpha_k d_k)^T d_k = 0$$

(Interpretation) This says the gradient at $x_{k+1} = x_k + \alpha_k d_k$ is orthogonal to the descent direction d_k .

Step 3: Orthogonality of consecutive gradients: Since $d_k = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$, the condition $\nabla f(x_{k+1})^T d_k = 0$ becomes:

$$\nabla f(x_{k+1})^T \left(-\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|} \right) = 0$$

Hence:

$$\nabla f(x_{k+1})^T \nabla f(x_k) = 0$$

(Key insight) Successive gradients are orthogonal! The algorithm zigzags.

Step 4: Take the limit: If $x_k \rightarrow x^*$ as $k \rightarrow \infty$, then by continuity of the gradient:

$$\nabla f(x_{k+1}) \rightarrow \nabla f(x^*)$$

and

$$\nabla f(x_k) \rightarrow \nabla f(x^*)$$

Step 5: Use orthogonality in the limit: From $\nabla f(x_{k+1})^T \nabla f(x_k) = 0$ for all k , taking the limit:

$$\nabla f(x^*)^T \nabla f(x^*) = 0$$

Step 6: Conclude:

$$\|\nabla f(x^*)\|^2 = 0 \implies \nabla f(x^*) = 0$$

If steepest descent with exact line search converges to x^* , then x^* is a stationary point.

Teaching Notes**Board Diagram****Draw the zigzag pattern:**

- Sketch 2–3 level curves of a convex function (elliptical contours)
- Show starting point x_0 far from optimum
- Draw the first descent direction d_0 (perpendicular to level curve at x_0)
- Follow this direction until tangent to a level curve (at x_1)
- From x_1 , draw d_1 perpendicular to the level curve
- Show that the path zigzags toward the optimum
- Label: “Consecutive gradients are orthogonal”
- Remark: For ill-conditioned problems, this zigzag is very slow!

Pedagogical emphasis: The orthogonality is the *defining feature* of exact line search. It guarantees stationarity at convergence but can lead to slow convergence.

Common Confusion

Student Mistake: “Doesn’t the exact line search guarantee we reach the optimum in a few steps?”

Answer: No! Exact line search guarantees orthogonality of consecutive gradients, which can lead to slow zigzagging (especially for ill-conditioned problems). Reaching stationarity requires the limit $x_k \rightarrow x^*$ to hold. We’ll see the convergence rate in Theorem 3.

3 Theorem 3: Global Convergence Rate for Quadratic

Statement

Consider the quadratic function $f(x) = \frac{1}{2}x^T Qx + c^T x$ where $Q \succ 0$ (positive definite). Using steepest descent with exact line search, the iterate x_{k+1} satisfies:

$$q(x_{k+1}) \leq q(x_k) \left(1 - \frac{\lambda_{\min}(Q)}{\lambda_{\max}(Q)}\right) = q(x_k) \left(1 - \frac{1}{\kappa(Q)}\right)$$

where $q(x) := f(x) - f(x^*)$ is the optimality gap, $\lambda_{\min}, \lambda_{\max}$ are the smallest and largest eigenvalues of Q , and $\kappa(Q) := \lambda_{\max}/\lambda_{\min}$ is the **condition number**.

Proof

Step 1: Optimum for quadratic: The minimizer is $x^* = -Q^{-1}c$, and:

$$f(x^*) = -\frac{1}{2}c^T Q^{-1}c$$

The optimality gap is:

$$q(x) = f(x) - f(x^*) = \frac{1}{2}(x - x^*)^T Q(x - x^*)$$

Step 2: Descent direction: The steepest descent direction is $d_k = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$. For the quadratic, $\nabla f(x_k) = Qx_k + c = Q(x_k - x^*)$ (using $c = -Qx^*$).

Step 3: Exact line search for quadratic: For a quadratic with direction d_k , the exact line search step size is:

$$\alpha_k = \frac{\|\nabla f(x_k)\|^2}{\nabla f(x_k)^T Q \nabla f(x_k)}$$

(Verification) With this choice, the directional derivative at $x_k + \alpha_k d_k$ is zero.

Step 4: Express the next iterate:

$$x_{k+1} = x_k + \alpha_k d_k = x_k - \alpha_k \frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$$

Shift to centered coordinates: $y_k := x_k - x^*$, so $\nabla f(x_k) = Qy_k$:

$$y_{k+1} = y_k - \frac{\|Qy_k\|^2}{\|Qy_k\|} \cdot \frac{1}{Qy_k^T Q \frac{Qy_k}{\|Qy_k\|}} \cdot \frac{Qy_k}{\|Qy_k\|}$$

After algebraic simplification:

$$y_{k+1} = y_k - \frac{\|Qy_k\|^2}{y_k^T Q^2 y_k} \cdot Qy_k$$

Step 5: Bound the gap: We have:

$$q(x_{k+1}) = \frac{1}{2} y_{k+1}^T Q y_{k+1}$$

Using the **Rayleigh quotient** bound: for any v ,

$$\lambda_{\min}(Q) \|v\|^2 \leq v^T Q v \leq \lambda_{\max}(Q) \|v\|^2$$

and properties of the exact line search, one can show:

$$q(x_{k+1}) \leq \frac{(\lambda_{\max} - \lambda_{\min})^2}{(\lambda_{\max} + \lambda_{\min})^2} \cdot q(x_k)$$

This simplifies to:

$$q(x_{k+1}) \leq \left(\frac{\kappa(Q) - 1}{\kappa(Q) + 1} \right)^2 q(x_k)$$

or equivalently (first-order approximation):

$$q(x_{k+1}) \leq \left(1 - \frac{1}{\kappa(Q)} \right) q(x_k)$$

The linear convergence rate is controlled by the condition number $\kappa(Q)$.

Teaching Notes

Insights and Discussion

Key takeaways:

- **Condition number determines speed:** If $\kappa(Q) = 1$ (spherical level sets), we converge in one step! If $\kappa(Q)$ is large (elongated ellipses), convergence is slow.
- **Number of iterations:** To reduce the gap by factor ϵ , we need $k \approx \kappa(Q) \log(1/\epsilon)$ iterations. For ill-conditioned problems ($\kappa \gg 1$), this is very slow.
- **Comparison:** Newton's method achieves quadratic convergence and doesn't depend on conditioning. This is a major advantage!

Board illustration:

- Left side: Draw elliptical level curves (elongated, high condition number)
- Overlay the zigzag steepest descent path (many iterations)
- Right side: Show spherical level curves (condition number = 1)
- Steepest descent converges immediately

Common Confusion

Student Mistake: “Why does the condition number matter for steepest descent but not for Newton’s method?”

Answer: Steepest descent only uses first-order information (the gradient). For an ill-conditioned quadratic, the gradient points in a direction that’s not aligned with the optimal descent path, forcing zigzagging. Newton’s method uses second-order information (the Hessian) to “unskew” the geometry, making the problem spherical locally. We’ll see this in Theorem 4.

4 Theorem 4: Newton Direction is a Descent Direction

Statement

If the Hessian $\nabla^2 f(x_k) \succ 0$ (positive definite) and $\nabla f(x_k) \neq 0$, then the Newton direction

$$d_k^N := -(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

is a descent direction, i.e., $\nabla f(x_k)^T d_k^N < 0$.

Proof

Step 1: Define the Newton direction:

$$d_k^N = -(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

Step 2: Compute the directional derivative:

$$\begin{aligned} \nabla f(x_k)^T d_k^N &= \nabla f(x_k)^T (-(\nabla^2 f(x_k))^{-1} \nabla f(x_k)) \\ &= -\nabla f(x_k)^T (\nabla^2 f(x_k))^{-1} \nabla f(x_k) \end{aligned}$$

Step 3: Use positive definiteness: Let $g := \nabla f(x_k) \neq 0$ and $H := \nabla^2 f(x_k) \succ 0$. Then:

$$\nabla f(x_k)^T d_k^N = -g^T H^{-1} g$$

Since $H \succ 0$, we have $H^{-1} \succ 0$. For any nonzero vector g :

$$g^T H^{-1} g > 0$$

$\Rightarrow$

$$\nabla f(x_k)^T d_k^N = -g^T H^{-1} g < 0$$

The Newton direction is a strict descent direction.

Teaching Notes

Geometric Interpretation

Affine scaling perspective:

The Newton direction can be understood as the steepest descent direction in a transformed metric induced by $H = \nabla^2 f(x_k)$:

$$d_k^N = -(\nabla^2 f(x_k))^{-1} \nabla f(x_k) = \text{steepest descent w.r.t. the metric } H$$

This “pulls” the gradient direction toward the “true” descent path by accounting for the local curvature.

Board diagram:

- Draw an elliptical level curve (ill-conditioned problem)
- Show the gradient $\nabla f(x_k)$ pointing perpendicular to the level curve
- Show the Newton direction d_k^N pointing more directly toward the optimum
- Annotate: “Newton direction accounts for curvature”

Common Confusion

Student Mistake: “What if $\nabla^2 f(x_k)$ is not positive definite?”

Answer: If the Hessian has negative eigenvalues (saddle point), then the Newton direction might point uphill instead of downhill. This is why the positive definiteness assumption is essential. In practice, algorithms use modified Newton methods or trust regions to handle indefinite Hessians.

5 Theorem 5: Newton's Method Converges in 1 Step for Quadratics

Statement

For a quadratic function $f(x) = \frac{1}{2}x^T Qx + c^T x$ where $Q \succ 0$, Newton's method reaches the optimum $x^* = -Q^{-1}c$ in exactly one step from any starting point x_0 :

$$x_1 = x^* \quad (\text{exact})$$

Proof

Step 1: Compute gradient and Hessian:

$$\nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q$$

Note that both are independent of x .

Step 2: Newton direction at x_0 :

$$d_0^N = -(\nabla^2 f(x_0))^{-1} \nabla f(x_0) = -Q^{-1}(Qx_0 + c)$$

Step 3: Newton update with unit step: Setting $\alpha = 1$ (Newton's method uses unit step):

$$\begin{aligned} x_1 &= x_0 + d_0^N = x_0 - Q^{-1}(Qx_0 + c) \\ &= x_0 - Q^{-1}Qx_0 - Q^{-1}c \\ &= x_0 - x_0 - Q^{-1}c \\ &= -Q^{-1}c = x^* \end{aligned}$$

Newton's method reaches the global optimum in one iteration for any quadratic function.

Teaching Notes

Comparison with Steepest Descent

Contrast with Theorem 3:

- **Steepest Descent (SD):** Linear convergence, requires $O(\kappa(Q) \log(1/\epsilon))$ iterations
- **Newton:** Exact convergence in 1 step!

Why the difference?

Steepest Descent uses only $\nabla f(x_k)$ (first-order info) and must “greedily” optimize along one direction at a time. Newton uses the full Hessian to find the exact Newton direction that directly points to the optimum.

Analogy:

- SD is like walking downhill on a misaligned path, making many turns
- Newton is like teleporting directly to the valley

Board diagram:

- Left: Zigzag path of SD (multiple iterations)
- Right: Straight arrow of Newton (one step to x^*)

Common Confusion

Student Mistake: “If Newton is so good, why don’t we always use it?”

Answer: Computing $(\nabla^2 f(x_k))^{-1}$ is expensive ($O(n^3)$ operations) and solving the linear system $\nabla^2 f(x_k) \cdot d = -\nabla f(x_k)$ requires a well-conditioned Hessian. For large-scale problems or when the Hessian is hard to compute, steepest descent or quasi-Newton methods are preferred.

6 Theorem 6: Quadratic Convergence of Newton's Method (Local)

Statement

Under suitable conditions near a stationary point x^* where $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$ (positive definite Hessian), Newton's method exhibits **quadratic convergence**:

$$\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2$$

for some constant $C > 0$ and sufficiently small initial error $\|x_0 - x^*\|$.

Key Ideas (Proof Sketch)

Step 1: Taylor expansion: For Newton's method, the update is $x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$.

Step 2: Error recursion: Let $e_k := x_k - x^*$ be the error. Using Taylor's theorem:

$$\nabla f(x_k) = \nabla f(x^* + e_k) = \nabla^2 f(x^*) e_k + O(\|e_k\|^2)$$

Step 3: Newton update with error:

$$e_{k+1} = e_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

Substituting the expansion and assuming $\nabla^2 f(x_k) \approx \nabla^2 f(x^*)$ when e_k is small:

$$e_{k+1} \approx -(\nabla^2 f(x^*))^{-1} [O(\|e_k\|^2)]$$

$\Rightarrow$

$$\|e_{k+1}\| = O(\|e_k\|^2)$$

Step 4: Formal statement: This is quadratic convergence. The constant C depends on the Hessian and higher derivatives of f near x^* .

Near the optimum, Newton's method converges much faster than linear methods. Each iteration r

Intuition and Rate Comparison

Convergence Rates:

- **Linear convergence (SD):** $\|e_k\| \leq C^k \|e_0\|$ for $C < 1$. Many iterations needed for high accuracy.
- **Quadratic convergence (Newton):** $\|e_k\| \leq M \|e_{k-1}\|^2$. Each iteration doubles the number of correct digits!

Example: If $\|e_0\| = 10^{-1}$, then:

- After 1 Newton step: $\|e_1\| \approx M \cdot 10^{-2}$ (2 correct digits)
- After 2 Newton steps: $\|e_2\| \approx M \cdot 10^{-4}$ (4 correct digits)
- After 3 Newton steps: $\|e_3\| \approx M \cdot 10^{-8}$ (8 correct digits)

Teaching Notes

Practical Remarks

Key distinctions:

- **“Local” convergence:** Quadratic convergence only holds for $\|x_0 - x^*\|$ sufficiently small. Far from x^* , Newton may diverge or stall.
- **“Global” convergence:** To guarantee convergence from any starting point, one uses line search (Newton with line search) or trust region methods.
- **Practical implication:** Use steepest descent initially (to get “close” to x^*), then switch to Newton for final iterations.

Sketch on board:

- Plot a convex function with a narrow valley
- Show steepest descent converging slowly from far away
- Show Newton converging very fast once close to x^*
- Label: “Quadratic convergence *near* the optimum”

Reference: For a complete proof, see Bazaraa, Sherali, and Shetty, *Nonlinear Programming*, Theorem 8.6.5 or similar graduate nonlinear optimization texts.

Common Confusion

Student Mistake: “If Newton converges in one step for quadratics, doesn’t that mean the convergence is quadratic everywhere?”

Answer: No. Quadratic convergence only applies to convergence near a stationary point (local behavior). For quadratics, Newton happens to *reach* the exact solution in one step, which is even stronger than quadratic convergence. Far from x^* , Newton can diverge or exhibit slow progress on non-quadratic functions.

Summary Table: Method Comparison

Property	Steepest Descent	Newton's Method	Remarks
Direction	$d_k = -\nabla f(x_k)$	$d_k = -(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$	Newton uses Hessian (curvature) info
Quadratic function	$O(\kappa \log(1/\epsilon))$ steps	1 step (exact)	Huge advantage for Newton
Convergence rate	Linear	Quadratic (local)	Newton much faster near x^*
Cost per step	$O(n^2)$ (+ line search)	$O(n^3)$ (Hessian inverse)	Trade-off: accuracy vs cost
Robustness	Global convergence (with LS)	Local convergence only	Need line search or trust region for Newton
Practical use	Large-scale, ill-conditioned	Medium-scale, well-conditioned	Often use both: SD then Newton

Lecture Script Outline (for instructor)

Opening (2 minutes)

[Today we'll study two fundamental optimization algorithms: Steepest Descent and Newton's Method. Both aim to minimize a differentiable function $f(x)$, but they use different information and have very different convergence properties.]

Theorem 1 & 2: Steepest Descent (12–17 minutes)

[We start with Steepest Descent (SD). The idea is to move in the direction that decreases f the most rapidly. We'll prove that this direction is $d^ = -\nabla f(x_k)/\|\nabla f(x_k)\|$.]*

Sketch Theorem 1 proof: Draw the unit ball, the gradient, and use Cauchy-Schwarz to show why the opposite gradient direction is optimal.

[Great. Now, suppose we perform an exact line search at each iteration. What can we say about convergence?]

Sketch Theorem 2 proof: Emphasize the orthogonality of successive gradients. Draw the zigzag pattern. This is the key geometric insight.

[Notice: SD does converge to a stationary point, but we haven't said anything about how fast. For that, we need to look at quadratic functions.]

Theorem 3: Rate Analysis (6–8 minutes)

[For a quadratic $f(x) = \frac{1}{2}x^T Qx + c^T x$, we can compute the convergence rate exactly. The answer? It depends on the condition number $\kappa(Q)$.]

State the result: Linear convergence with rate $1 - 1/\kappa(Q)$.

[If κ is large (ill-conditioned), SD is slow. This is a major practical limitation!]

Theorems 4 & 5: Newton's Method (10–12 minutes)

[Enter Newton's Method. Instead of just the gradient, we'll use the Hessian matrix to find a better direction.]

Sketch Theorem 4 proof: Show that the Newton direction is a descent direction (when Hessian is PD).

[Now for the killer result: on a quadratic function, Newton reaches the optimum in exactly one step. Regardless of the condition number!]

Sketch Theorem 5 proof: Direct calculation. Simple but powerful.

[Compare this to SD, which needs $\kappa(Q)$ iterations. This shows why Newton is so valuable for ill-conditioned problems.]

Theorem 6: Quadratic Convergence (6–8 minutes)

[Finally, for general (non-quadratic) functions, Newton exhibits quadratic convergence near the optimum. This means the number of correct digits roughly doubles each iteration.]

[But there's a catch: this fast convergence is only local. Far from x^ , Newton might diverge. In practice, we often use SD to get close, then switch to Newton for the final fast phase.]*

Closing (2 minutes)

[To summarize: SD is globally robust but slowly convergent; Newton is superfast near the optimum but requires Hessian computation and has local convergence. Understanding both is essential for modern optimization.]